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Brain Drain Detector

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Project Goal & Dataset

Multimodal detection of cognitive overload

Goal

- **Detect cognitive overload** in dyadic debates.
- **Per 5 s window:** Safe vs Alarm.
- **Fuse three modalities** (speech + physiology + EEG).

Dataset — K-EmoCon

- **16 sessions · 32 participants.**
- **Audio · E4 wristband · EEG** (synchronized).
- **Self-annotations** (arousal & valence).

Reframes emotion recognition as **conversational overload detection**.

Dataset Structure

Modalities, labels and the processed window

Speech Audio

- **Stereo WAV** → 16 kHz.
- **Diarization** (2 speakers).
- **VAD** 5 s windows.

Physiology — E4

- **EDA · HR · IBI.**
- **Wristband sensors.**
- **+ ACC, BVP, TEMP.**

Brain — EEG

- **NeuroSky, 1 channel.**
- **Theta · Alpha · Beta.**
- **Aligned to 10 Hz.**

One 5-second window — what the model receives

P1 | 5 s

Audio waveform

80,000 samples @ 16 kHz

Biosignals tensor (50 × 6)

50 steps
6 channels

Ground Truth

Binary labels from self-reported arousal & valence

Data strategy

- **Source:** P{N}.self.csv only.
- **Arousal & Valence (1–5);** emotion flags ignored.
- **One label** per 5 s window.

Classification rule

- **Safe (0):** Valence > 3 OR Arousal < 4.
- **Alarm (1):** Valence ≤ 3 AND Arousal ≥ 4.
- **Thresholds** set in configs/base.yaml.

Binary classifier: **Safe** = operational stability, **Alarm** = critical cognitive overload.

Audio Preprocessing

From raw stereo debate to clean 5-second windows

Signal Processing

- **Split stereo WAV** into left / right channels.
- **Diarization** maps speakers to P_left / P_right.
- **Silence masking** of the competing speaker.
- **Fill short gaps** (< 3 s) inside a turn.
- **Resample** all tracks to 16 kHz.

VAD & Windowing

- **PyAnnote VAD** on each separated track.
- **Fixed 5 s windows** (consecutive).
- **Keep a window** if speech ≥ 3 s or ≥ 60% overlap.
- **Export** to audio/*.pt as float32 tensors.

Physio Preprocessing

Six channels aligned into one biosignal tensor

Signal Mapping

- **E4 wristband:** EDA, HR, IBI.
- **NeuroSky EEG:** Theta, Alpha, Beta.
- **Tensor order:** [EDA, HR, IBI, theta, alpha, beta].
- **Excluded:** ACC, BVP, TEMP.

Alignment & Resampling

- **5 s consecutive windows.**
- **Linear interpolation** to sync sample rates.
- **50 steps / window** (10 Hz grid).
- **Z-score** per window, per channel.

Output: `{"biosignals": tensor(50, 6), "participant": "P1", "seconds": 5}`

Weighted Loss: Balanced Alarm Detection

Correcting model bias from class imbalance

The imbalance problem

- **84% Safe** vs 16% Alarm.
- **Model drifts** to “always Safe”.

Weighted solution

- **Inverse-frequency** class weights.
- **Recomputed** per LOSO fold.

① Count classes → ② $w_c = N / (2 \cdot count_c)$ → ③ Weighted CrossEntropy

$$w_c = \frac{N}{2 \cdot count_c}$$

Example · 400 Safe / 100 Alarm

Safe	0.625	*low penalty*
Alarm	2.500	*high penalty*

→ an Alarm miss costs **4× more** to the model.

Balanced Participant Sampling

Correcting bias caused by unequal window counts per participant.

The scaling problem

- **Unequal counts** (40 vs 70 windows).
- **Talkative subjects** dominate gradients.
- **Risk:** overfit to few participants.

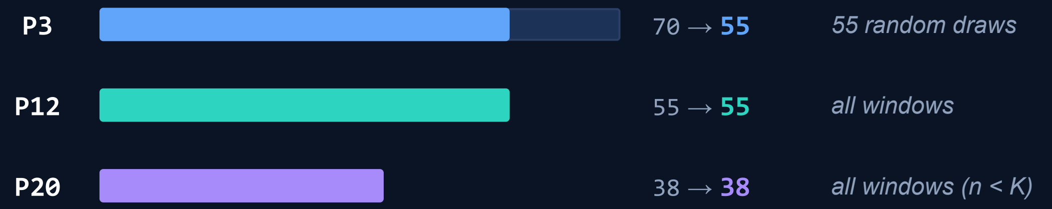
Sampling logic

- **K = median** (n_i) per LOSO fold.
- **Draw min(n_i, K)** per participant.
- **Shuffle** into balanced epoch.

Per-epoch goal

- **Equal cap** on each participant.
- **Short sessions:** all windows.
- **Long sessions:** random K subset.

Example · K = median = 55



→ Each participant contributes at most **K windows** per epoch.

Audio Embedding

Two encoder backends, fused with biosignals

Input

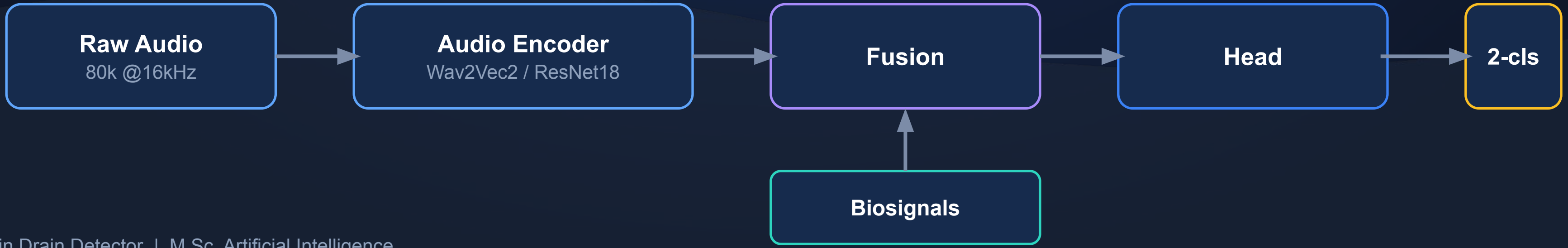
- **Raw waveform** 80,000 floats @ 16 kHz.

Fusion

- **Cross-attention** with biosignals → 256-D → head.

Encoder Options

Wav2Vec2
pretrained, frozen → mean pool → 768-D.



Biosignal Embedding

A BiGRU summarizes the physiological window

Input

- **6 signals × 50 steps** (EDA, HR, IBI + 3 EEG).

BiGRU Encoder

- **Bidirectional GRU;** efficient on medium data.

Pooled (default)

One 256-D vector (128 × 2 final states).

Sequence

50 × 256-D → cross-attention weights moments in time.

6 × 50 tensor → **BiGRU** → **Embedding(s)** → **Fusion**

«Balanced» Cross-Attention

Fusing audio and biosignals without an audio bias

The problem

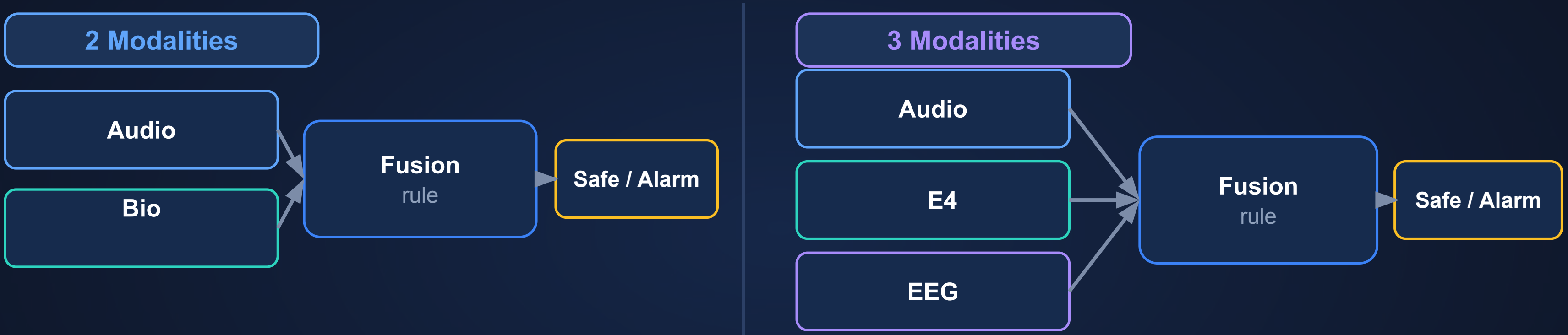
- **Default residual = audio only** → audio-heavy.
- **Pooled token (len 1):** softmax always 1.

Our recipe

- **Balanced residual:** learn w_a, w_b on both.
- **Modality dropout (p=0.15):** forces both branches.

Late (Decision-Level) Fusion

Combine independently trained unimodal models



Uniform avg
mean of probabilities

Val-F1 weighted
weight by validation F1

Majority / OR
alarm if any fires

Stacking (LR)
meta-classifier on probs

Quality weighted
weight by signal quality

Same 5 methods · weights fit per LOSO fold · tested on both 2-mod and 3-mod configurations.

Late Fusion — Results

Pooled LOSO · synthetic 50/50 confusion matrix · 2 vs 3 modalities

Metric (%)	Quality-weighted		Stacking LR		Majority vote (OR)	
	2-mod	3-mod	2-mod	3-mod	2-mod	3-mod
Accuracy	51.9	52.3	59.3	**62.3**	59.0	59.6
Recall · Alarm	18.4	14.2	47.6	**51.6**	48.4	48.4
F1 · Alarm	27.7	22.9	53.9	**57.8**	54.1	54.5
Precision · Alarm	55.8	59.7	62.1	**65.7**	61.4	62.4
Macro-F1	45.8	44.2	58.7	**61.9**	58.5	59.1
Recall · Safe	85.4	**90.4**	71.0	73.0	69.6	70.8
Precision · Safe	51.1	51.3	57.5	**60.1**	57.4	57.8
F1 · Safe	64.0	65.5	63.6	**65.9**	62.9	63.7
Specificity · Safe	85.4	**90.4**	71.0	73.0	69.6	70.8

- **Stacking LR (boxed)** is the best 2-mod and 3-mod combiner.
- **+3 mod (EEG):** +3.0 acc, +3.1 macro-F1 over 2-mod.
- **Quality-weighted** collapses to Safe (Alarm recall 14–18%).
- **Majority-OR** close, but trails stacking everywhere.

Late Fusion vs. Cross-Attention Baseline

Best late-fusion combiner against the single end-to-end model

Metric (%)	Late fusion Stacking LR (2-mod)	Cross-attn balanced, no aug
Accuracy	**59.3**	48.3
Recall · Alarm	**47.6**	12.0
F1 · Alarm	**53.9**	18.8
Precision · Alarm	**62.1**	43.8
Macro-F1	**58.7**	40.5
Recall · Safe	71.0	**84.6**
Precision · Safe	**57.5**	49.0
F1 · Safe	**63.6**	62.1
Specificity · Safe	71.0	**84.6**

What the numbers say

- **+18.3 macro-F1,** +35.6 Alarm recall.
- **Baseline looks safe** only by over-predicting Safe.
- **Fusion actually** catches the minority class.

Selection by mean rank over all 9 metrics: **Stacking 2.22** < Majority 2.78 < Quality 2.89 < Uniform 3.44 < Val-F1 3.67

Cross-Attention Fusion: Pooled & Sequence

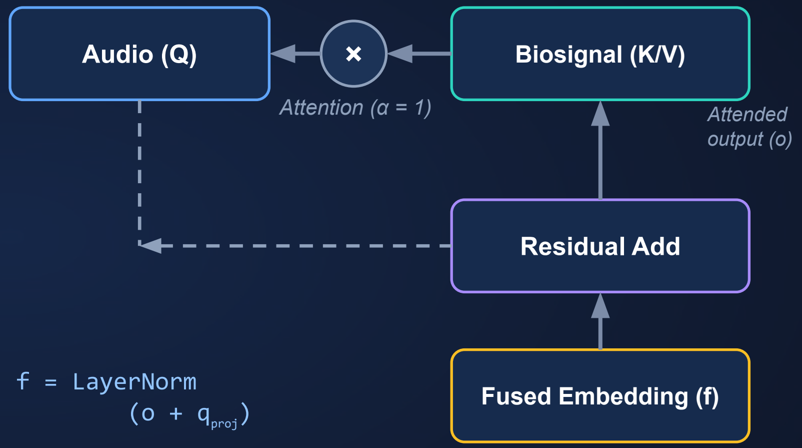
Audio-anchored attention over biosignals — two token modes

How it works

- **Query = audio** embedding (the anchor).
- **Keys / Values =** biosignals (context).
- **4-head attention** + residual + LayerNorm.

Two token modes

- **Pooled** (shown): 1 token → α ≡ 1, o = v.
- **Sequence:** 50 time tokens → attention over time.



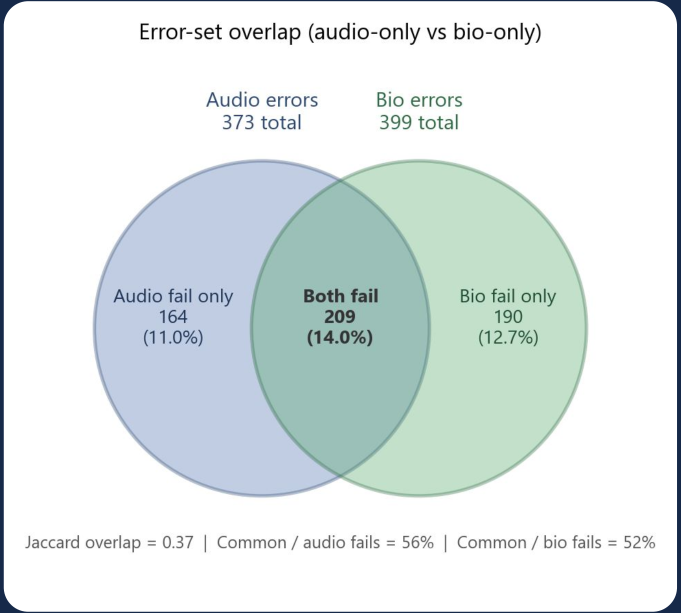
Difference from the «balanced» cross-attention

Balanced adds learned residual weights (w_a, w_b) + modality dropout; **pooled** keeps a single audio-anchored residual — fewer parameters, **less overfitting on this small dataset.**

Are the Modalities Complementary?

Where audio-only and bio-only models fail (pooled LOSO test windows)

Error-set overlap



Reading the overlap

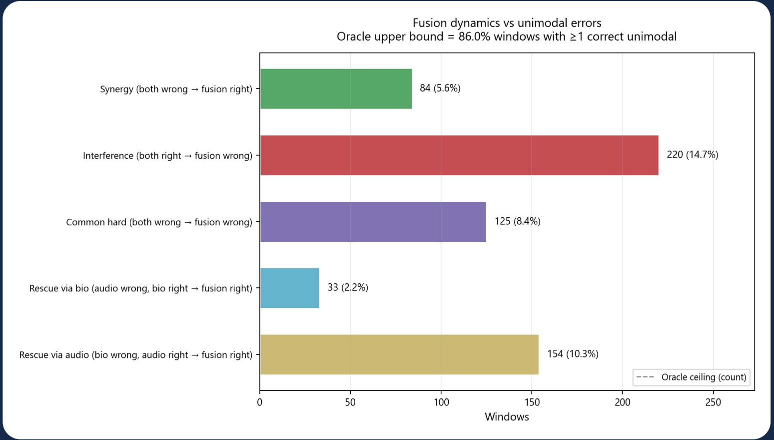
- **Different mistakes:** audio & bio errors barely overlap (Jaccard 0.37).
- **Only 14% both-fail** — 11% audio-only, 12.7% bio-only.
- **Each modality** rescues windows the other misses.

Oracle (pick the right modality) = **86% accuracy** → large headroom to fuse.

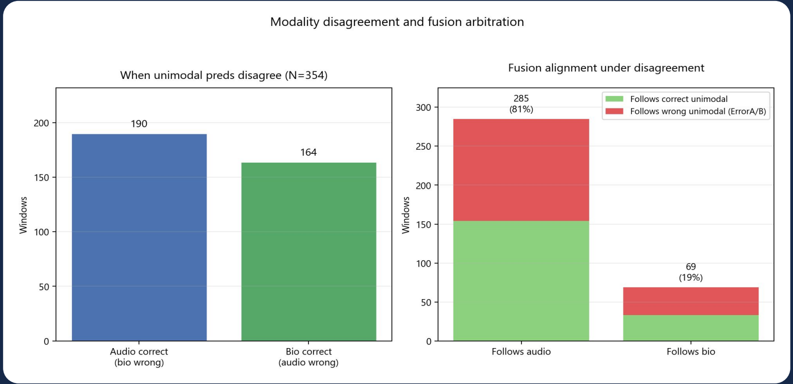
Why Static Fusion Underperforms

Fusion dynamics and modality arbitration under disagreement

Fusion vs. unimodal errors



Who does fusion follow?



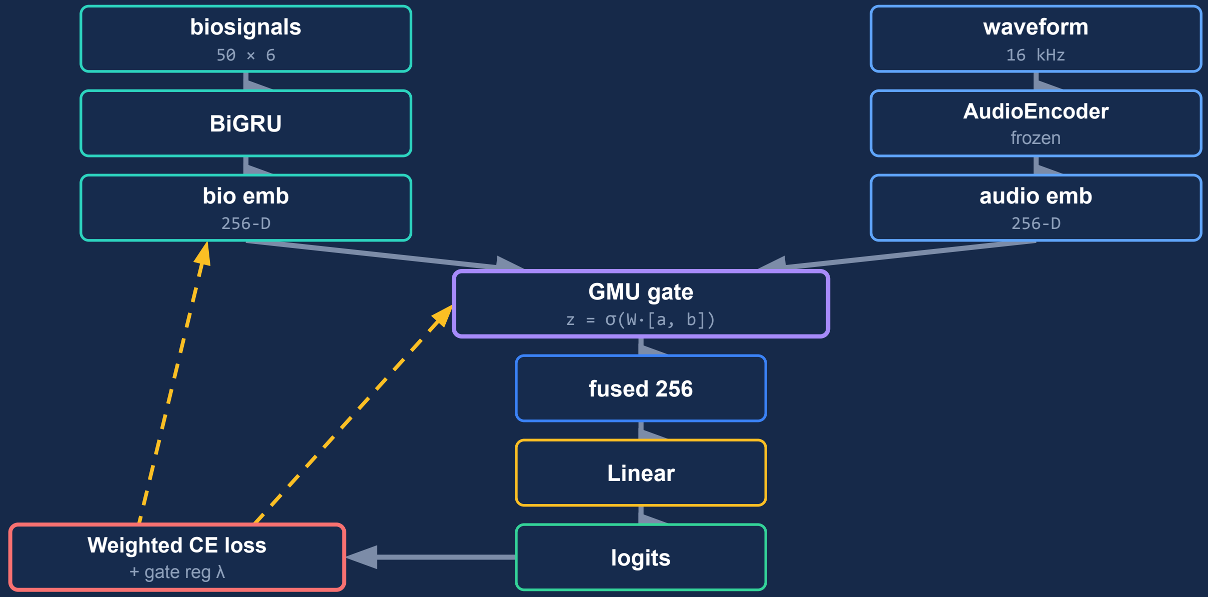
- **Interference (220):** both right → fusion wrong.
- **Audio rescues 154** windows vs. only 33 for bio.
- **Under disagreement,** fusion follows audio 81%.
- → **Need a combiner** that learns when to trust each modality.

Gated Multimodal Unit (GMU)

Computational graph — forward pass & gradient flow

Computational graph

solid → forward · **dashed → gradient (loss)**



Fusion Models — Head-to-Head

Balanced 50/50 metrics · cross-attn block vs. late fusion vs. GMU

Metric (%)	Cross-attention				Late fusion		GMU
	Cross-attn balanced	Cross-attn pooled ★	Cross-attn sequence	Cross-attn unweighted	Late fusion Stacking 2-mod	Late fusion Stacking 3-mod ★	GMU gated fusion
Accuracy	48.3	59.8	57.0	49.8	59.3	**62.3**	52.0
Recall · Alarm	12.0	51.2	43.4	1.6	47.6	**51.6**	21.2
F1 · Alarm	18.8	56.0	50.2	3.1	53.9	**57.8**	30.6
Precision · Alarm	43.8	61.8	59.6	44.4	62.1	**65.7**	55.2
Macro-F1	40.5	59.5	56.2	34.6	58.7	**61.9**	47.0
Recall · Safe	84.6	68.4	70.6	**98.0**	71.0	73.0	82.8
Precision · Safe	49.0	58.4	55.5	49.9	57.5	**60.1**	51.2
F1 · Safe	62.1	63.0	62.1	**66.1**	63.6	65.9	63.3
Specificity · Safe	84.6	68.4	70.6	**98.0**	71.0	73.0	82.8

- **Pooled cross-attn ★** best early fusion (59.50% Macro-F1).
- **Stacking 3-mod ★** best overall system (61.86% Macro-F1).
- **Unweighted pooled:** 1.6% Alarm recall — no class weights.
- **Balanced & GMU lag:** extra mechanism hurts on small data.

Dual-Signals Biosignal Encoder

Separate E4 and EEG towers — only the biosignal encoder changes

Hypothesis

- **Two devices,** distinct physiology.
- **E4:** EDA, HR, IBI (autonomic / stress).
- **EEG:** θ, α, β (brain activity).
- **Idea:** per-device encoders may beat a joint BiGRU.

Experimental setup

- **Input:** 5 s window (50 × 6).
- **Baseline:** single BiGRU (h=128) for all 6 channels.
- **Ablation:** dual towers (h=64 each) → concat.
- **Fair test:** both output a 256-d vector.

Architectural flow

** Only the biosignal encoder is modified; audio (Wav2Vec2) stays frozen.*

Dual-Signals Encoder — Results

Balanced 50/50 metrics · does splitting E4 and EEG help?

Metric (%)	Cross-attn dual-signal	Cross-attn pooled (1 BiGRU)	Late fusion 3-mod stacking
Accuracy	52.1	59.8	**62.3**
Recall · Alarm	24.2	51.2	**51.6**
F1 · Alarm	33.6	56.0	**57.8**
Precision · Alarm	54.8	61.8	**65.7**
Macro-F1	48.1	59.5	**61.9**
Recall · Safe	**80.0**	68.4	73.0
Precision · Safe	51.4	58.4	**60.1**
F1 · Safe	62.5	63.0	**65.9**
Specificity · Safe	**80.0**	68.4	73.0

Verdict

- **Dual towers lose** to the single BiGRU (48 vs 60 macro-F1).
- **Splitting by device** adds parameters, no signal gain.
- **Other E4+EEG merges** (e.g. cross-attention) were also worse.
- **Single pooled encoder** stays the simpler, stronger choice.

Ablation: 1D CNN + BiGRU for Biosignals

Local temporal features before the recurrent encoder

Physio 50 × 6 → **1D CNN** K=5, 6→32 ch → **BiGRU** h=128 → **256-d** → **Fusion** + audio

Metric (%)	BiGRU single (baseline)	1D CNN + BiGRU CNN front-end
Accuracy	**59.8**	51.1
Recall · Alarm	**51.2**	20.8
F1 · Alarm	**56.0**	29.8
Precision · Alarm	**61.8**	52.8
Macro-F1	**59.5**	46.2
Recall · Safe	68.4	**81.4**
Precision · Safe	**58.4**	50.7
F1 · Safe	**63.0**	62.5
Specificity · Safe	68.4	**81.4**

Why it didn't help

- **Goal:** catch sharp local patterns (EDA spikes) before the GRU.
- **Result:** macro-F1 drops 59.5 → 46.2.
- **CNN front-end** over-smooths / overfits short noisy windows.
- **Plain BiGRU** generalizes better here.

Ablation: Inter-Window GRU / LSTM ×5

Architecture — causal context over five fused embeddings per participant

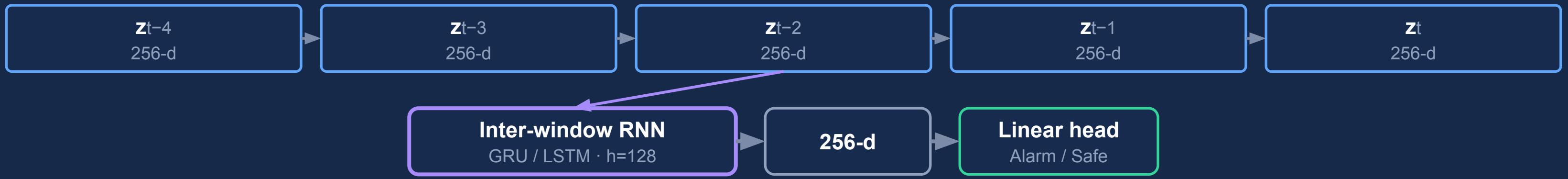
What changes

- **Baseline:** one pooled cross-attn vector z per 5 s window.
- **Ablation:** stack the last 5 z vectors per participant.
- **RNN:** causal GRU or LSTM (h=128) over the sequence.
- **Output:** final hidden state → linear Alarm / Safe head.

Hypothesis

- **Overload** may build over ~25 s of debate speech.
- **Each z** fuses audio + physio for one VAD window.
- **Sequence** follows debate time (gaps allowed).
- **Expectation:** temporal context should beat single-window pooling.

Inter-window sequence model



zt = pooled cross-attention fusion per window · encoders fixed, only head changes.

Ablation: Inter-Window GRU / LSTM ×5

Results — temporal context hurts vs. single-window pooled baseline

$z_{t-4\dots t}$ 5 × 256-d → **Inter-window** GRU / LSTM → **256-d** → **vs.** pooled z → **Alarm** / Safe

Metric (%)	LSTM ×5 windows	GRU ×5 windows	Pooled single window
Accuracy	51.1	56.7	**59.8**
Recall · Alarm	18.4	29.2	**51.2**
F1 · Alarm	27.3	40.3	**56.0**
Precision · Alarm	53.2	**64.9**	61.8
Macro-F1	45.2	53.2	**59.5**
Recall · Safe	83.8	**84.2**	68.4
Precision · Safe	50.7	54.3	**58.4**
F1 · Safe	63.1	**66.0**	63.0
Specificity · Safe	83.8	**84.2**	68.4

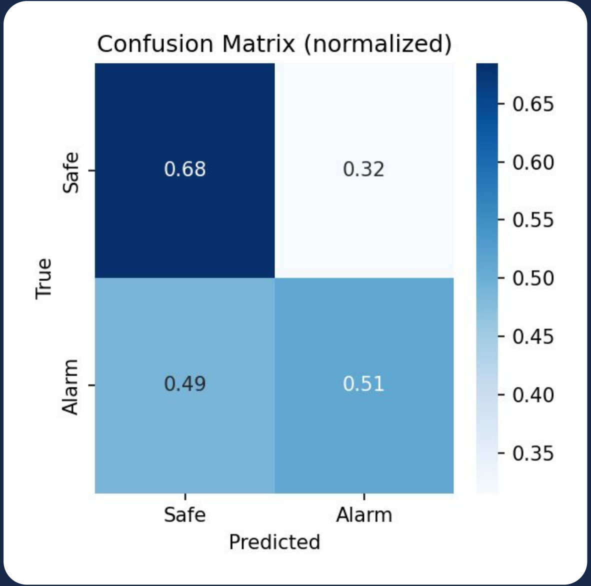
Why it didn't help

- **Goal:** overload builds over ~25 s of speech (5 windows).
- **Result:** Macro-F1 drops to 53.2 (GRU) / 45.3 (LSTM).
- **Alarm recall:** −22 pp (GRU) / −33 pp (LSTM) vs pooled.
- **VAD gaps** break contiguity — inter-window context adds noise.
- **Takeaway:** single-window pooled fusion remains the better early model.

Best Run — Pooled Cross-Attention

Confusion matrix and balanced-vs-real metrics (n = 1492)

Confusion matrix (norm.)



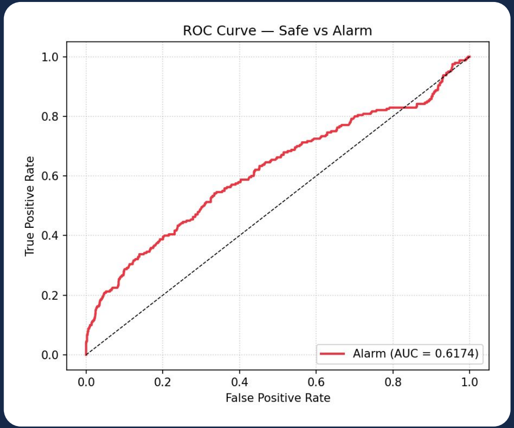
Metric (%)	Balanced 50/50	Real class split	Δ (pp) real − bal
Accuracy	59.8	65.7	5.9
Recall · Alarm	51.2	51.2	0.1
F1 · Alarm	56.0	32.5	-23.6
Precision · Alarm	61.8	23.8	-38.1
Macro-F1	59.5	54.7	-4.8
Recall · Safe	68.4	68.5	0.1
Precision · Safe	58.4	88.0	29.6
F1 · Safe	63.0	77.0	14.0
Specificity · Safe	68.4	68.5	0.1

On the real imbalanced split accuracy rises, but Alarm precision & F1 collapse — the minority class stays genuinely hard.

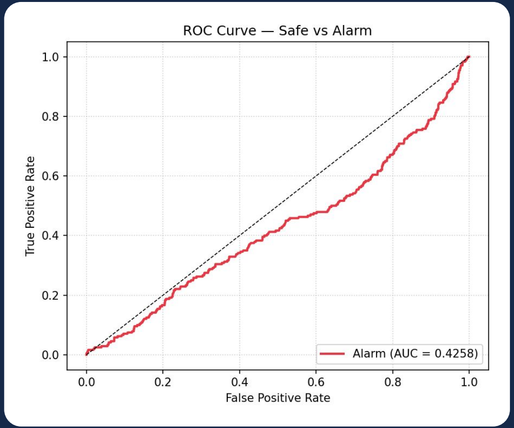
ROC — Cross-Attention Ablation

Pooled cross-attention · weighted vs. unweighted loss + sampling

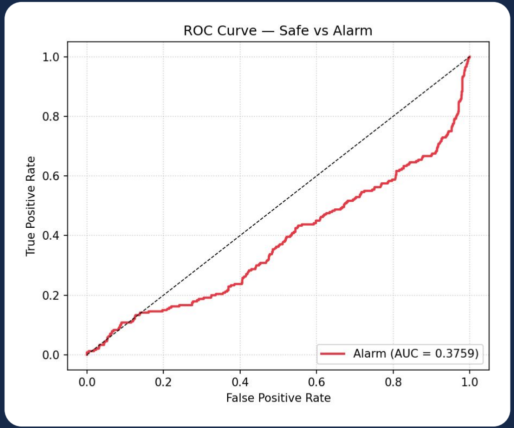
Pooled · weighted



Balanced cross-attn



Pooled · unweighted

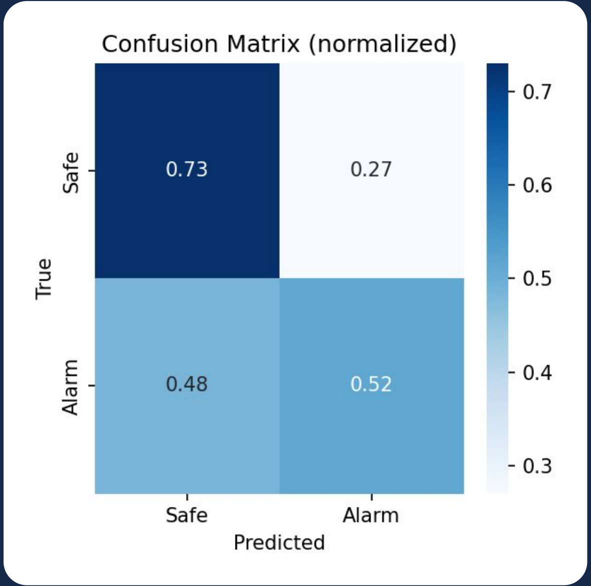


Weighted + balanced sampling: **51.2% Alarm recall** · Unweighted: **1.6% recall** — curve collapses toward Safe majority.

Best Run — Late Fusion (Stacking LR, 3 mod)

Confusion matrix and balanced-vs-real metrics (n = 1492)

Confusion matrix (norm.)



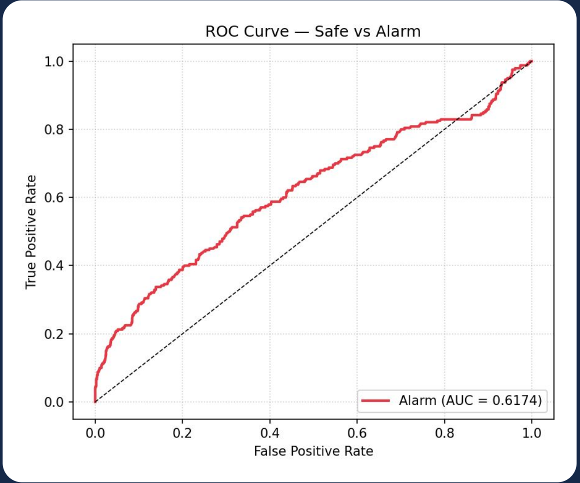
Metric (%)	Balanced 50/50	Real class split	Δ (pp) real − bal
Accuracy	62.3	69.5	7.2
Recall · Alarm	51.6	51.7	0.1
F1 · Alarm	57.8	35.3	-22.5
Precision · Alarm	65.7	26.8	-38.9
Macro-F1	61.9	57.7	-4.2
Recall · Safe	73.0	72.9	-0.1
Precision · Safe	60.1	88.7	28.6
F1 · Safe	65.9	80.0	14.1
Specificity · Safe	73.0	72.9	-0.1

Best overall system (61.86% Macro-F1). Alarm recall is stable on the real split (+0.07 pp); precision/F1 drop as in early fusion.

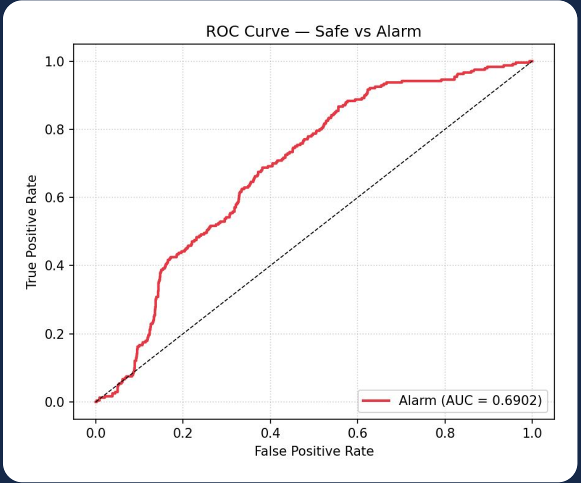
ROC — Best Early vs. Best Late Fusion

Side-by-side on pooled LOSO test predictions (real class distribution)

Best early fusion · pooled cross-attn



Best late fusion · stacking LR (3 mod)



Late fusion ranks Alarm windows slightly better across thresholds (**61.86%** vs **59.50%** Macro-F1) — matches the best overall system.

What We Learned

Honest takeaways — including what did not work

What worked

- **Simple beats complex:** pooled cross-attn & stacking late fusion won.
- **Class weighting** restored minority (Alarm) recall.
- **Complementary errors:** oracle 86% → real fusion headroom.

What didn't — and why

- **Balanced residual + dropout:** over-regularized, broke the audio anchor.
- **GMU & dual-signal towers:** extra parameters, no gain on small data.
- **Sequence attn & CNN front-end:** fit noise instead of signal.

K-EmoCon is small, noisy and weakly labelled (self-reported affect in dyadic debates) — prior work reports near-chance to modest scores, in line with our AUC ≈ 0.6.

Thank you

Questions?

Brain Drain Detector · Multimodal Cognitive-Overload Detection